IsDB
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Closing The Technology Gaps:

Empowering Women in The Rice Postproduction Value Chain



2025

Acknowledgements

This report is a unique example of what we can achieve when we work together. It exemplifies how effective partnership can drive impact within the development arena. It is indeed pleasing that the Islamic Development Bank (IsDB) secured a grant of USD 11.25 Million from the Women Entrepreneurs Finance Initiative (We-Fi) to implement the Program on Empowering West African Women Small and Medium Enterprises in Rice Value Chains (EWASME) in four West African countries - Guinea, Niger, Senegal and Sierra Leone.

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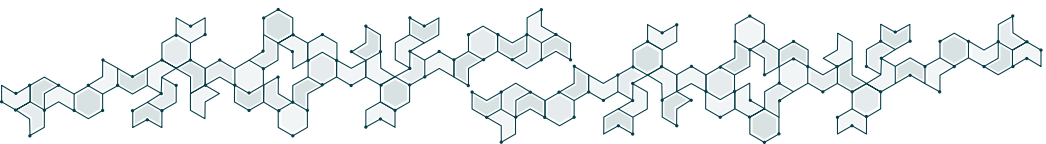
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LIST OF ABBREVIATIONS

ECOWAS	Economic Commission of West African States
EWASME	Empowering West African Women SMEs in the Rice Value Chains
FAO	Food and Agriculture Organization of the United Nations
ICT	Information and Communications Technology
IsDB	Islamic Development Bank
SMEs	Small and Medium Enterprises
VSEs	Very Small Enterprises
We-Fi	Women Entrepreneurs Finance Initiative
WSMEs	Women Small and Medium Enterprises
WVSEs	Women Very Small Enterprises



Foreword



Rice is more than a staple crop in West Africa—it is a foundation of food security, a driver of rural livelihoods, and a strategic commodity with the potential to transform economies. Yet, behind the millions of tons produced annually lies a persistent and often overlooked challenge: the technological divide that limits women's full participation in the rice postproduction value chain. This report, Closing the Technology Gaps: Empowering Women in the Rice Postproduction Value Chain, is both a diagnostic and a call to action. It offers a timely and rigorous analysis of the constraints faced by women entrepreneurs and enterprises in accessing and utilizing postharvest technologies, and it charts a path toward inclusive, sustainable transformation.

The Islamic Development Bank (IsDB), in partnership with the Women Entrepreneurs Finance Initiative (We-Fi), is proud to support the Empowering West African Women Small and Medium Enterprises in Rice Value Chains (EWASME) Program. This initiative spans Guinea, Niger, Senegal, and Sierra Leone—four countries where rice is not only a dietary staple but also a symbol of resilience and opportunity. Through EWASME, we aim to strengthen women-led Very Small Enterprises (WVSEs) and Small and Medium Enterprises (WSMEs) by enhancing their access to finance, technology, and markets. This report is a critical output of that effort, synthesizing insights from country-level studies and desk research to illuminate the barriers and opportunities that define women's engagement in rice postproduction.

This report is the second in the series. The first series looked at the binding constraints that women entrepreneurs in rice post-production face and identified applicable solutions to the challenges women entrepreneurs encounter with respect to access to finance. Despite their central role in postharvest activities—threshing, drying, milling, parboiling, and marketing—women entrepreneurs remain marginalized in terms of access to mechanized equipment, digital tools, and infrastructure. The reasons are multifaceted: limited financial capacity, gender-biased technology design, sociocultural norms, and inadequate training. These constraints not only reduce the efficiency and profitability of women-led enterprises but also perpetuate gender disparities in agricultural value chains. The report estimates postproduction losses of up to 22% among WVSEs and WSMEs—losses that could be significantly reduced through targeted technological interventions.

Yet, the report does more than diagnose problems. It offers a roadmap for change. It calls for a holistic, ecosystem-based approach to rice postproduction development—one that integrates gender-responsive policies, inclusive financing mechanisms, capacity building, and infrastructure investment. It highlights the importance of promoting women-friendly technologies such as the GEM parboiling system, and it underscores the need for digital literacy and access to mobile banking and e-commerce platforms. Crucially, it advocates for co-financing models and sustainable partnerships that empower women not just as beneficiaries, but as agents of transformation.

This report is also a testament to collaboration. It draws on the expertise of researchers, practitioners, and policymakers across the region. The synthesis of country-level data from Guinea, Niger, Senegal, and Sierra Leone provides a nuanced understanding of local contexts while revealing common threads that run across borders. The contributions of national consultants, technical specialists, and institutional leaders have enriched the analysis and ensured its relevance to both policy and practice.

As we look ahead, we must recognize that closing the technology gap is not merely a technical challenge—it is a matter of equity, empowerment, and economic resilience. Women in rice postproduction are not just laborers or traders; they are innovators, leaders, and entrepreneurs. By investing in their potential, we invest in the future of food systems, rural economies, and inclusive development.

We hope this report will serve as a valuable resource for governments, development partners, financial institutions, and civil society organizations committed to advancing gender equity and agricultural transformation. Let it inspire new partnerships, inform policy reforms, and catalyze investments that unlock the full potential of women in the rice value chain.

Dr. Issa Faye,
Director General,
Global Practice and Partnerships





Executive **Summary**

The Women Entrepreneurs Finance Initiative (We-Fi) is supporting IsDB's efforts to implement the "Empowering West African Women SMEs (EWASME) program in the Rice Value Chains", which has a total budget of USD 11.25 million. The program aims to support women entrepreneurs in four major rice-producing countries in West Africa (Guinea, Niger, Senegal, and Sierra Leone) to better engage in the rice value chain, particularly in the postproduction stage. The program addresses three main areas: (i) financial constraints; (ii) non-financial constraints; and (iii) poor links between women entrepreneurs and existing private sector-led value chains that could address constraints in input and output markets.

Rice is a strategic crop that provides an exit from poverty and food insecurity. West Africa is the largest rice-producing region in the continent, producing a little over 10 million tons annually. Rice production provides millions of smallholder farmers in the region with jobs, sources of income, and livelihoods. The gross value of rice produced in West Africa has been increasing consistently, hitting almost USD 10 billion in 2022.

Women entrepreneurs are involved in various scales along the value chain. Countries have varying definitions of very small, small, and medium size enterprises. The definitions may also vary across sectors within the same country. However, a common thread across the definitions, countries, and sectors includes size, location, wage labor, target market, and sophistication of inputs. The majority of enterprises in West Africa are in the very small, small, and medium size categories.





Women's access to postproduction technologies is constrained. Access by WVSEs and WSMEs to rice postproduction technologies can reduce burdens, improve the physical quality of rice, and save time, which can then be used for household or other productive activities. However, such access and utilization are likely to be impeded by the lack of technical skills and knowledge, the complexity of new rice postproduction machines and equipment, and the inability to operate and maintain them effectively. Therefore, governments and development partners, in collaboration with rice postproduction machine and equipment suppliers, should organize relevant awareness campaigns and training for WVSEs and WSMEs.

Acquiring rice postproduction technologies requires high investments. This is a major barrier for women WVSEs and WSMEs, as they are usually constrained by resources. In this respect, governments and development partners, including financial institutions, should combine financial incentives with technical assistance, subsidies, grants, and/or low-interest loans to WVSEs and WSMEs to help them purchase or rent rice postproduction machines and equipment or pay providers of such services.

Improving access for WVSEs and WSMEs should not be done in isolation. Rather, it should be done within a framework of improving national digital infrastructure. At the enterprise level, there are basic conditions that need to be addressed. These include availability and affordability of technology, internet connectivity and stability, and availability of trained and skilled ICT staff. Equally important is the availability, stability, and affordability of electricity to power digital equipment and businesses.

WSMEs lose prominence as technologies and markets become more sophisticated. WSMEs have higher potential for the postproduction process transition towards higher and modern technologies and markets. It is at this stage that WSMEs could play a bigger role in connecting different components along the rice value chain. In addition, as rice postproduction processes, especially milling, become more commercialized and industrialized, SMEs play greater roles in linking other segments



Access by WVSEs and WSMEs to rice postproduction technologies can reduce burdens, improve the physical quality of rice, and save time, which can then be used for household or other productive activities.



of the value chain from production and processing to market. However, the prominence of women ownership declines as businesses become larger, more formalized, and use more sophisticated technologies.

WVSEs and WSMEs face barriers that limit their ability to leverage technology for business growth and innovation. The barriers to access of women entrepreneurs to these technologies are as follows: (i) difficulty to access finance for rice postproduction technologies; (ii) lack of technical knowledge and skills; (iii) lack of women-friendly technologies; (iv) insufficient skill; (v) market barriers; (vi) limited infrastructure; (vii) sociocultural norms; and (viii) training gaps.

Main training needs for WVSEs and SMEs. Training on the operation of postproduction machines is crucial, particularly the use and operation of threshers and winnowing machines, and the use and operation of small-scale rice-milling machines. Because machines are traditionally operated by men, women are usually not included in the training organized by suppliers when machines are installed. As postproduction activities in the rice value chain are dominated by women, it is only logical that women should be trained on the use, operation, maintenance, and management of postproduction machines.

Recommendations provided by this study should not be implemented in isolation. Also, noting the pervasive gender gaps in West Africa, especially in rural areas, special initiatives for women entrepreneurs are required to facilitate accessibility, affordability, and utilization of rice postproduction technologies. However, while a holistic approach is required to address barriers to WVSE and WSME access to technology and use, it must be done within the national gender and women economic empowerment policies.

Limitation of the study. While this study has largely focused on access of WVSEs and WSMEs to rice postproduction technologies, it did not assess the profitability of rice postproduction technologies or the conditions under which such technologies could be profitable or not profitable. However, it is a known fact that profitability of technology investments, especially by VSEs and SMEs, is dependent on several factors, including appropriateness of the technology, location, market demand, capacity, utilization per annum, incentives, training and skills, and product market.







01

Context and **Background**

The Women Entrepreneurs Finance Initiative (We-Fi) supports women entrepreneurs by scaling up access to financial products and services, building capacity, expanding networks, offering mentorship, and providing opportunities to link with domestic and global markets. The global economy is presently confronted with the most daunting and complex set of challenges and uncertainties, such as climate change, fragility, conflict, and food insecurity. These challenges and uncertainties cannot be addressed without engaging and empowering women, especially those in developing countries. Among the strategies to stimulate economic growth and promote women economic empowerment and gender parity, governments and international organizations have realized that investing in women-led businesses has proven to be one of the most effective strategies in creating inclusive growth and resilience. Accordingly, We-Fi estimates that gender parity in entrepreneurship could add USD 5-6 trillion in net value to the global economy – and each women-led firm is six times more likely to hire female employees than men-led ones.

According to its 2023 Annual Report, We-Fi has allocated, since its launch in 2017, USD 364 million to support women-led enterprises, out of which USD 166 million (46%) was directed to Sub-Saharan Africa[1]. Globally, more than 28,000 WSMEs have benefited from training opportunities provided by We-Fi, and 42 legal and regulatory reforms have been facilitated with We-Fi support. The Islamic Development Bank (IsDB) is among the implementing partners[2] of We-Fi and is implementing a 5-year program "Empowering West African Women-SMEs in the Rice Value Chains" (EWASME), with a total budget of USD 11.25 million. The program aims to support women entrepreneurs in four major rice-producing countries in West Africa (Guinea, Niger, Senegal and Sierra Leone) to better engage in the rice value chain, particularly at the postproduction stage. The program addresses three main areas: (i) financial constraints; (ii) non-financial constraints; and (iii) poor links between women entrepreneurs and existing private sector-led value chains that could address constraints in input and output markets.



1. We-Fi Annual Report, 2023
2. We-Fi implementing partners include the Asian Development Bank (ADB), the African Development Bank (AfDB), the European Bank for Reconstruction and Development (EBRD), the Inter-American Development Bank Group (IDBG), the Islamic Development Bank (IsDB), and the World Bank Group (WBG)



The estimated population of West Africa as of 2024 is approximately 456.25 million people, out of which 226.20 million are female, representing 49.58% of the total population[3]. A breakdown of the population by country shows that about five countries in West Africa have more females than men (Nzezkor, 2023) and contribute significantly to employment and entrepreneurship. However, more women than men tend to operate in the informal sector and make a significant contribution to the economy, and to the agricultural sector in particular. With respect to the agricultural sector in West Africa, women account for 80% of employment in processing, 70% of distribution, and almost 90% of sales of ready-to-consume agricultural products (Allen et al., 2018).

Women play important roles as entrepreneurs and in cross-border trade in West Africa. However, despite the important roles they play and their contribution to the economy, women entrepreneurs are confronted with distinct bottlenecks that inhibit the growth of their businesses, including limited access to finance, infrastructure, and technology. Over the years, West African governments and development partners have recognized the importance of women entrepreneurs in socioeconomic development and are putting in place support mechanisms to alleviate the constraints they face. One such initiative is the "Empowering West African Women SMEs in the Rice Value Chains" (EWASME) program that is being implemented by IsDB, with financial support from the We-Fi.

The EWASME program aims to support women entrepreneurs in four major rice-producing countries in West Africa (Guinea, Niger, Senegal and Sierra Leone) to better engage in the rice value chain, particularly in the postproduction stage. The program addresses three main areas: (i) financial constraints; (ii) non-financial constraints; and (iii) poor links between women entrepreneurs and existing private sector-led value chains that could address constraints in input and output markets. This report was prepared with assistance from IsDB to provide knowledge to help key stakeholders in supporting WVSE and WSME value-chain finance initiatives, in addition to the implementation of EWASME.

3. World Population Prospects (2024 Revision) - United Nations population estimates and projections. https://population.un.org/wpp/

1.1 Definitions of Key Concepts

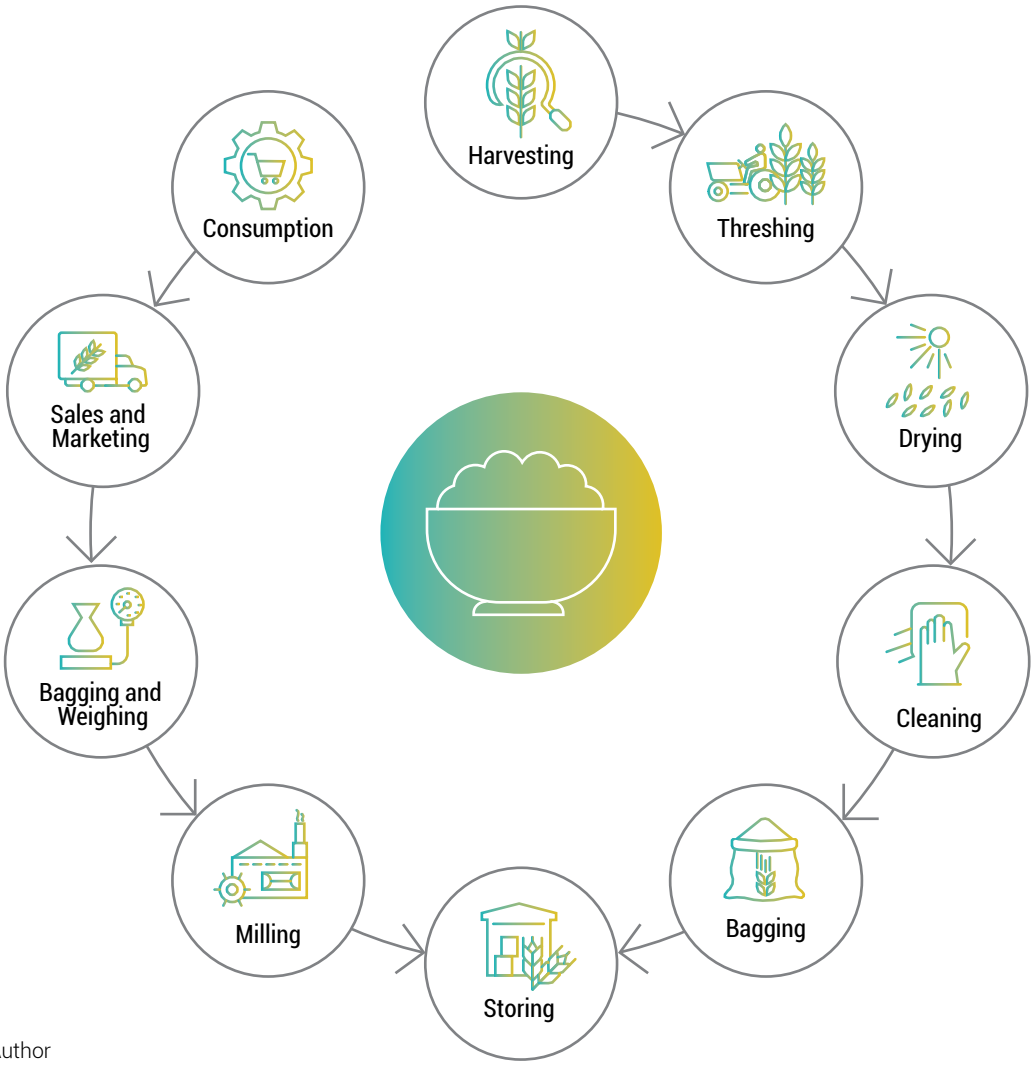
Rice Value Chain: The term "rice value chain" describes the full range of value-adding activities that firms/enterprises, farmers, and workers carry out to bring the product (rice) from its conception to its end use and beyond. The benefits of supplying specific value chains vary dramatically, depending on how these chains are governed; whether suppliers receive inputs, knowledge, and ancillary services as part of their engagement; and the targeted end market.

Rice Postproduction System: This consists of a set of operations that cover the phases from harvest to consumption. An efficient postproduction system aims to minimize losses and maintain the quality of the crop (or product) until it reaches the final consumer.

Rice Postproduction Process: This includes a series of processes and handling techniques that are part of the rice cultivation. They are applied from the harvest from the field for the purposes of transformation into a form, condition, or composition that adds value, makes it storable or prolongs its shelf-life, and makes it consumable or usable in other value-added products (see Figure 1).

Transport: Following its harvest, the rice is transferred from the field to a location for further handing and processing.

Figure 1: Rice Post-production processes



Source: Author



Threshing: Is the process of removing the paddy (husk rice) from the panicle for further processing. There are three methods for threshing rice: **rubbing**, **impact**, and **stripping**. During threshing, the paddy is detached from the panicle. Women carry out this manual process by trampling the panicles. Where mechanical threshers are available, the impact method is used.



Drying: At the time of harvest, the moisture content of the grains ranges from 24% - 26%. Drying reduces the moisture content of the paddy to 12-14% for safe storage and milling, or at most 18% for temporary storage of 2 weeks when it is not possible to dry any faster.



Paddy Cleaning/Winnowing: This involves the separation of undesirable material, such as weed seeds, straw, chaff, panicle stems, empty grains, immature and damaged grains, sand, rocks, stone, dust, plastic, and even metal and glass particles.



Storage: This involves the safekeeping of paddy or milled rice for the seed and food, respectively. At the farm household level, storage is essential for food security or as a commodity bank for conversion into cash when required.



Parboiling: Paddy rice to be milled as parboiled rice is completely soaked in containers (metal drums, large pots, etc.) between 12 and 24 hours, usually by women. The water is drained to about 20%, the container is covered, and the paddy rice is steamed over fire.



Milling: This operation involves the separation of the husk and the bran (polishing) to produce the edible portion for consumption. Rice mills vary from a manually operated beam hammer pounder or mortar and pestle to the sophisticated machines used in big commercial or government installations. In rural areas without energy, the beam hammer pounder or the mortar and pestle are used by farmers. Mortar and pestle is the most common rice milling method used by rural women. Women entrepreneurs who purchase paddy rice transport the paddy to commercial mills.



Women-owned Enterprises in Rice Postproduction: The definition of women-owned enterprise is provided in Box 1. The definition considers the ownership structure (owned by a majority of women) and level of investment (51% or more). It also considers the proportion (20% or more) of women in senior-management positions.



Postproduction Value Chain Finance: This is funding provided to or by a value-chain actor to increase value-chain growth and competitiveness. In practice, value chain finance can be provided by a bank; a buyer; or an input, machine, or equipment supplier. Value chain financing allows enterprises or firms to operate and transact with others and to upgrade. It involves providing financial services and products to one or more points in a value chain to reduce the risk and cost of financing and to increase the efficiency and competitiveness of the value chain.

Box 1: Definition of Women-owned Enterprise.

Woman-Owned Enterprise

An enterprise qualifies as a woman-owned enterprise if it meets the following criteria:

Criteria 1:

The enterprise is owned by 51% or more by a woman or women. In other words, the 51% of the investment or more is owned by a woman or women.

Or

Criteria 2:

The enterprise is owned by 20% or more by a woman or women and has one (1) woman or more serving as Chief Executive Officer (CEO), Chief Operating Officer (COO), President, or Vice President.

AND

Has 30% or more women serving as Board of Directors, where a board exist.

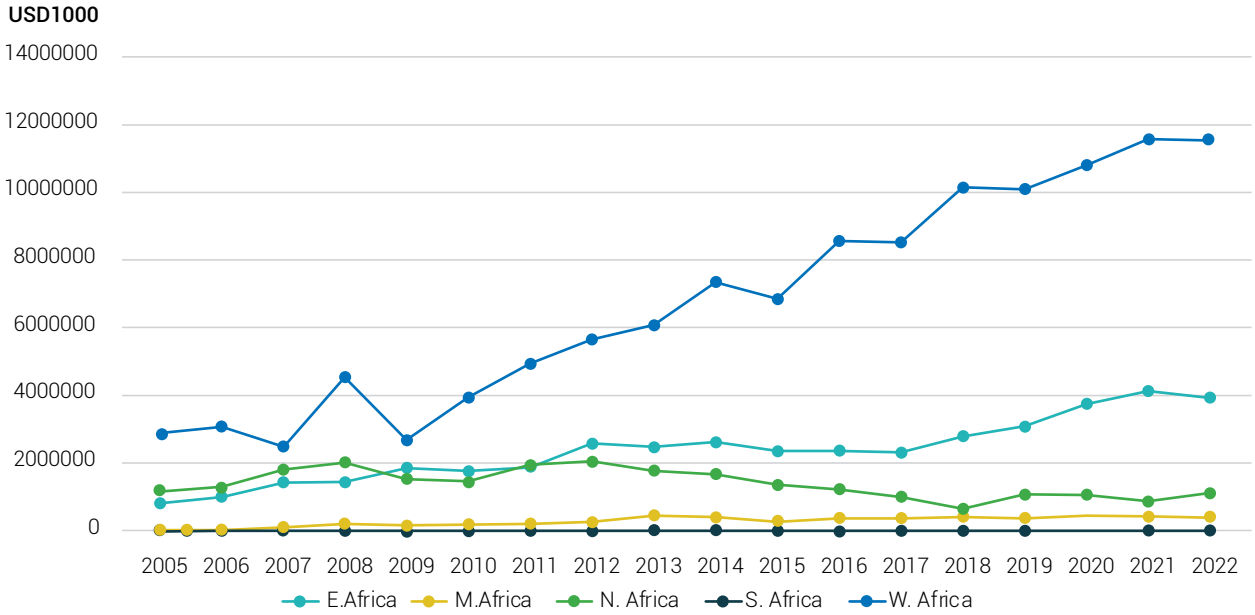
1.2 Rice in West Africa



West Africa is the largest rice-producing region in the continent, producing a little over 10 million tons annually. Rice production provides millions of smallholder farmers in the region with jobs, sources of income, and livelihoods. Thus, the rice sector provides an exit from poverty and food insecurity in the region, as rice availability and prices have become major determinants of the welfare of the poorest segments of African consumers. Figure 2 shows the gross production value of rice in each of the five regions of Africa. In West Africa, it has been increasing consistently, hitting almost USD10 billion in 2022. East Africa is the next most important region in terms of gross production value, with about USD 4 billion in the same year. While rice is produced in the other regions, their gross production value remains at USD 2 billion and tends to be declining.

Although rice production and gross production value have been increasing in West Africa over the past two decades, consumption has been increasing faster than production. This increase has been driven mainly by the growing population, urbanization, and changes in consumer preferences. According to Africa Rice, the demand for rice

Figure 2: Gross Production Value 2005-2022 (current USD 1000)



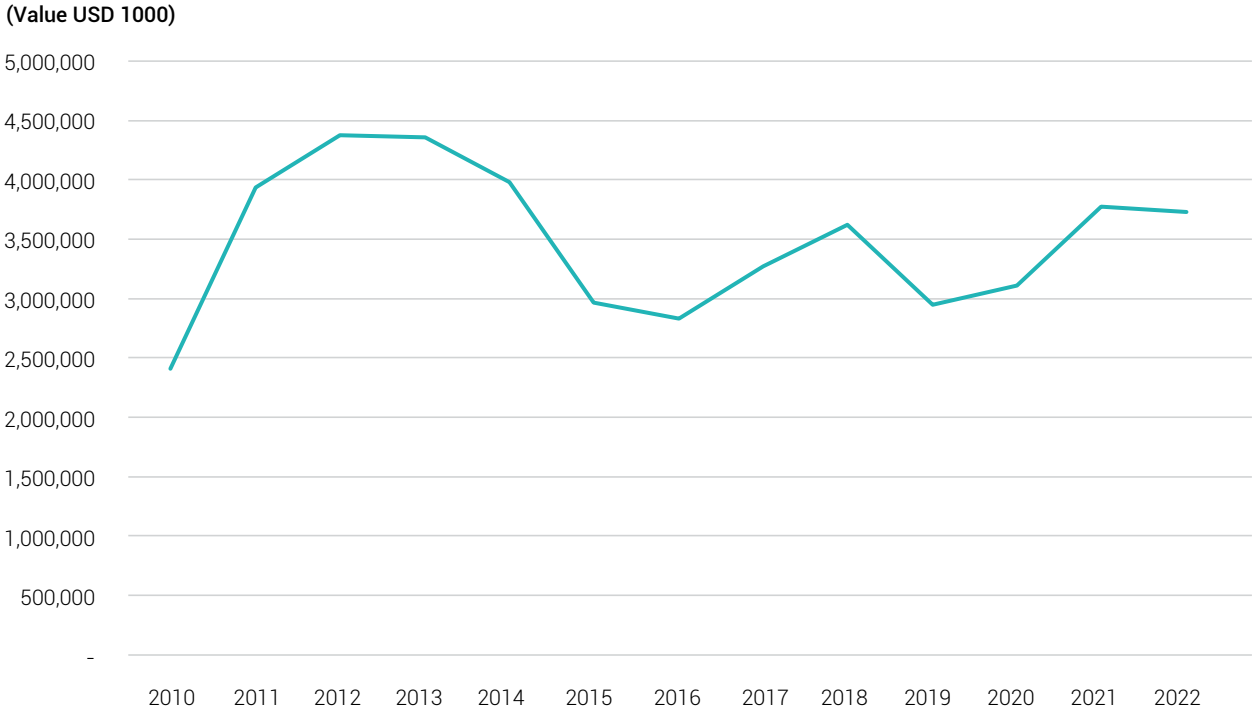
Source: Author, based on FAOSTAT.

is growing at more than 6% per year, which is faster than for any other food staple in sub-Saharan Africa. Data from FAO shows that Guinea, Guinea-Bissau, Liberia, and Sierra Leone have the highest per capita rice consumption (more than 90 kg per capita per year), followed by Senegal, Benin, Côte d'Ivoire, the Gambia, and Mali (with more than 50 kg per capita per year). According to FAO, rice consumption in West Africa is expected to grow by 70% to 24 million metric tons by 2025.

As a strategic commodity that can be used to tackle food insecurity in West Africa, countries have developed national rice sector development strategies. At the regional level, ECOWAS is playing a key role in supporting member states through regional input and market development policies to address the binding constraints to rice production in the region. West Africa has continued to face a structural rice deficit over the years, spending over USD 4 billion annually on rice imports (Figure 3). FAO has projected that the structural rice deficit will continue to widen in the years ahead, thus making the region increasingly reliant on imports.

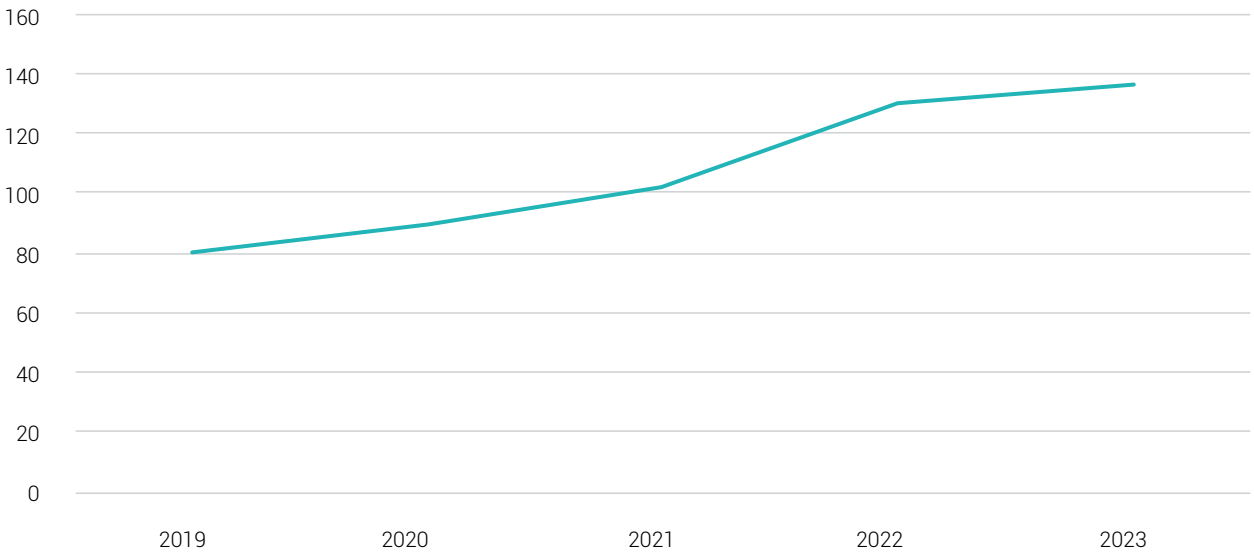
West Africa has continued to face a structural rice deficit over the years, spending over USD 4 billion annually on rice imports. FAO has projected that the structural rice deficit will continue to widen in the years ahead, thus making the region increasingly reliant on imports.

Figure 3: Rice Import Value in West Africa 2010-2022



Source: Author, based on FAOSTATS

Figure 4: International Rice Export Price Index



Source: Author, based on FAOSTAT

Figure 4 shows the international rice export price index. Prices have been increasing over the past 12 years. FAO projects that international rice prices will continue to rise due to trade policies of some exporting countries and adverse weather conditions that could impact production in major exporting countries. While the increasing rice price will hurt consumers, it provided opportunities for rice value chain actors in West Africa to increase production, processing, and trade to bridge the supply-demand gap, which is currently filled with imports.



1.3 Constraints to the Rice Value Chain Process

There are several constraints hindering rice production, widening the gap between supply from the region and the increasing consumption. The ECOWAS Rice factbook summarizes the main constraints in production, processing, and end use in West Africa. The key constraints are presented in Figure 5. Key production constraints include limited irrigated land area, use of low-level technologies, and low productivity. In the area of processing, limited access, use of inefficient technologies, insufficient supply of paddy for higher capacity utilization of mills, poor access to electricity, and limited use of byproducts feature as key bottlenecks. There are limited byproducts, leading to lower margins compared to regions where, in addition to milled rice, the sale of byproducts increases the margins of milling companies.

While women play important roles as entrepreneurs at each segment of the rice value chain, the constraints affect them more than their male counterparts, who have access to more resources and options at the household level. As shown by the World Bank, women entrepreneurs earn 34% less profit margin than men entrepreneurs across sub-Saharan Africa. This is attributed to gender discrimination, lack of a support network, and other social and self-limiting factors[4].

Figure 5: Key Constraints in the Rice Value Chain Process

Production	Processing	End Use
• Inadequate irrigation leads to highly volatile irrigated production is less than 5% of the 233,500 HA of land suitable for irrigation productivity levels. • Low levels of farming technologies mostly rudimentary for subsistence. • Yields hampered by inability to mitigate impact of weeds, insects, and birds	• Traditional manual processing and outdated equipment of mill processors generate high physical and quality losses of grains. • Bigger mills have difficulty sourcing sufficient • quantities of good quality paddy to maintain their facilities at full capacity. • Power failures increases cost dramatically. • Limited use of by-products like husks and straws due to lack of appropriate technology	• Local produce passes through several middlemen, squeezing margins for retailers. • Perceived high quality of imports because of cleanliness and packaging leads consumers to shun local rice. • Fragmented supply dominated by smallholders with low bargaining power. • Selling on the open market is the dominant retail distribution mode.

Source: ECOWAS Rice Fact Book

4. World Bank (n.d). Profiting from Parity: Unlocking the Potential of Women's Businesses in Africa. https://documents.worldbank.org/en/publication/documents-reports/documentdetail/501971553025918098/main-report



1.4 Women in the Rice Value Chain in West Africa

Both women and men play important and distinct roles in the rice value chain in West Africa. The roles and responsibilities of men and women in the rice value chain are dynamic and can change over time, depending on the rice-growing ecology. In general, men are mainly involved in land preparation (brushing, ploughing, and levelling) and transportation, while women play important roles in production and postproduction activities. The majority of rice farmers are smallholders who depend on family labor and some hired labor on a seasonal basis. Women are often involved in rice-farming activities that are mostly performed manually, such as sowing, transplanting, weeding, harvesting, and post-harvesting activities, including threshing, winnowing, parboiling, drying, and milling.

In general, women play a major role in the rice value chain. It is estimated that they make up 70% of the agricultural workforce and produce 80% of the crops (UNWOMEN 2021). However, women face several challenges in agriculture. Most notably, women have little access to land, education, and finance compared to men. Despite these limitations, women play major roles in the various economic activities along the rice value chain, which are vital not only for ensuring food security and poverty reduction, but also for job creation.

In the postproduction segment of the rice value chain, women are mainly engaged in lower-skilled activities that are generally manual. This is due to their limited skills and know-how, and their lack of access to the capital needed to purchase equipment, productive resources, and enough rice to supply within the value chain.

Postproduction activities are mostly carried out by women using basic technologies. This is further exacerbated by limited incentive for improving access to postproduction technologies, leading to poor-quality milled rice and low prices and profitability of domestic rice. Poor postproduction processes reduce both the quantity and quality of rice available for consumption, therefore undermining all four pillars of food security - availability, access, utilization, and stability. Thus, postproduction management is an important step in ensuring the profitability of rice farming and household and community food security, as well as upgrading WVSEs and WSMEs. In Sierra Leone, and in specific rice-consuming communities in Guinea,

Mali, and Senegal, where food security is synonymous to rice self-sufficiency, a community that is rice self-sufficient is resilient to external shocks and threats. Good postproduction management can bolster that resilience.

While there are several improved rice postproduction machines and equipment available on the market, women face gender barriers to access and adoption. Specifically, women are constrained to adopt certain postproduction technologies due to several factors, such as the design of technology, low incomes, and poor access to credit, land, and other physical resources. Women in rice postproduction activities are also constrained by limited access to information, as well as intra-household barriers. Cultural factors, such as limited participation of women in household decision-making and other gender-based role differentiations at the household level, affect equal access to agricultural technologies.

Although women dominate the rice retail market, they operate at a micro level and are confined to roles in the periphery of the rice distribution network, which is dominated by men. This situation makes it difficult for women to access necessary resources to upgrade their enterprises and to play leading roles in the postharvest segments of the rice value chain, which would have enabled them to maximize their generated wealth and connect with key value chain actors to expand their business opportunities.

Women in rice postproduction activities are also constrained by limited access to information, as well as intra-household barriers. Cultural factors, such as limited participation of women in household decision-making and other gender-based role differentiations at the household level, affect equal access to agricultural technologies.





1.5 Objectives of the Report

The purpose of this report is to provide data and information that could be used to upgrade and advance WVSEs and WSMEs, particularly at the postproduction stage, through increased access to finance and technical assistance. That would in turn enhance their access to and use of rice postproduction technologies, especially machinery and equipment, digital technology, energy, and market for milled rice, thus improving their competitiveness and growth.

The report was prepared with financial support from IsDB under the Women Entrepreneurs Finance Initiative (We-Fi) – a financial intermediary fund established to support WVSEs and WSMEs in developing countries. The program aims to stimulate increased support from the public and private sectors to create an enabling environment and open new doors for women entrepreneurs across the developing world, as well as to unlock financing for WVSEs and WSMEs in developing countries.

1.6 Methodology

This report is based on the country reports from Guinea, Niger, Senegal, and Sierra Leone that were prepared from commissioned studies by the IsDB EWASME program. The preparation of this report involved reviewing and synthesizing the four country reports and complementing them with data and information from published sources. Data and information from the country reports were complemented by desk research to enrich the thematic content.

The desk research was conducted in two stages. First, related works on women owned/led SMEs in general, and in West Africa in particular, were reviewed to identify common threads between women owned/led VSEs and SMEs. This was followed by a literature review on women entrepreneurs' access to technology in general and to rice postproduction technologies in particular. The International Rice Research Institute (IRRI) knowledge products on rice value chain development were reviewed and relevant information was extracted to complement data and information from the reports of the commissioned research in Guinea, Mali, Senegal, and Sierra Leone. In terms of technology, the data and information, and thus this report, focus on rice postproduction machinery and equipment and the use of digital technology among WVSEs and WSMEs in rice postproduction.





02

Women Enterprises and Access to **Technology**

2.1 Characteristics of SMEs and VSEs

Women entrepreneurs are involved in various scales along the value chain. While the size factor bears significance, countries have varying definitions of what makes an enterprise micro, small, or medium. The definitions may also vary across sectors within the same country. However, a common thread across the definitions, countries, and sectors includes size, location, wage labor, target market, and sophistication of inputs. Given this backdrop, the characteristics of these groups of enterprises, as provided by the International Finance Cooperation (IFC), are summarized in Table 1.

Table 1: Definition of Micro, Small, and Medium Enterprises

IFC MSME Definition				Loan Size Proxy
Indicator	Employees	Total Assets USD	Annual Sales USD	Loan Size at Origination
Micro	< 10	< 100,000	< 100,000	<USD 10,000
Small	10-49	100,000 - < 3 million	100,000 - < 3 million	<USD 100,000
Medium	50-300	3 million - 15 million	3 million - 15 million	<USD 1 or USD 2 million

Source: IFC Definition of targeted sectors

IFC has provided a standardized definition of MSMEs. Important variables used to define MSMEs include the value of owned fixed assets, annual sales, and the number of wage-earning employees. In addition to size, the location of these enterprises also matters in the context of West Africa.

Size is the most frequently used variable in defining MSMEs. In describing MSMEs, IsDB adopts the definition provided by IFC. Therefore, a ***microenterprise*** is a business with a total asset base less than USD 100,000 and annual sales less than USD 100,000. ***Small enterprises*** have a total asset base of USD 100,000 to less than USD 3 million. Medium enterprises have a total asset base of USD 3 million to USD 15 million. In the rice postproduction value chain in West Africa, women dominate the small enterprise category and their ownership or participation decreases as the size of the business increases.

Location is also an important classification variable. In West Africa, microenterprises and small enterprises in the rice postproduction value chain are often operated from homes, on the roadside, or have no fixed address. Medium enterprises may have known addresses, but mostly operate informally.

Labor defines the type of enterprise. Most small and microenterprises use family labor or informal



IsDB adopts the definition provided by IFC. Therefore, a microenterprise is a business with a total asset base less than USD 100,000 and annual sales less than USD 100,000. Small enterprises have a total asset base of USD 100,000 to less than USD 3 million. Medium enterprises have a total asset base of USD 3 million to USD 15 million.



apprentices and pay minimal or in-kind wages. In principle, a microenterprise has between 5 and 10 paid workers, while small enterprises have 10 to 49 paid workers, and medium enterprises have between 50 and 300 workers.

Produce for domestic market: Micro, small, and medium enterprises in West Africa produce goods and services that target the domestic market, which is subject to high competition and volatile demand. In some cases, they create backward and forward linkages within the country's economy by using locally produced inputs as well as produced inputs for larger industries. In terms of technological sophistication, they use labor intensive technologies that are less sophisticated and costly than their larger counterparts. The characteristics of micro, small, and medium sized enterprises and their implications for women are listed in Table 2.

Table 2: Characteristics of VSE and SMEs and Implications for Women

	Structure	Constraints	Skills/ Knowledge	Location	Hiring Practices
Characteristics of very small, small, and medium enterprises	- Flexible - Loosely structured - Not present in all sectors - Underperforming	- Less education, training, and experience are required - Lower level of capitalization required - Use of family labor rather than wage labor	- Can use already existing skills - Limited skills - Acquire skills by doing	- Street location - Homebased	Women dominate as owners of certain businesses
Implications for women	Enable women to divide their time between domestic responsibilities and economically productive activities	Because there are few barriers to entry into the micro and small enterprise sector, women can easily participate	Traditional activities such as garment making and food processing can be turned into businesses	Women can be involved in several activities at a time and overhead cost is low	Women entrepreneurs tend to employ women employees



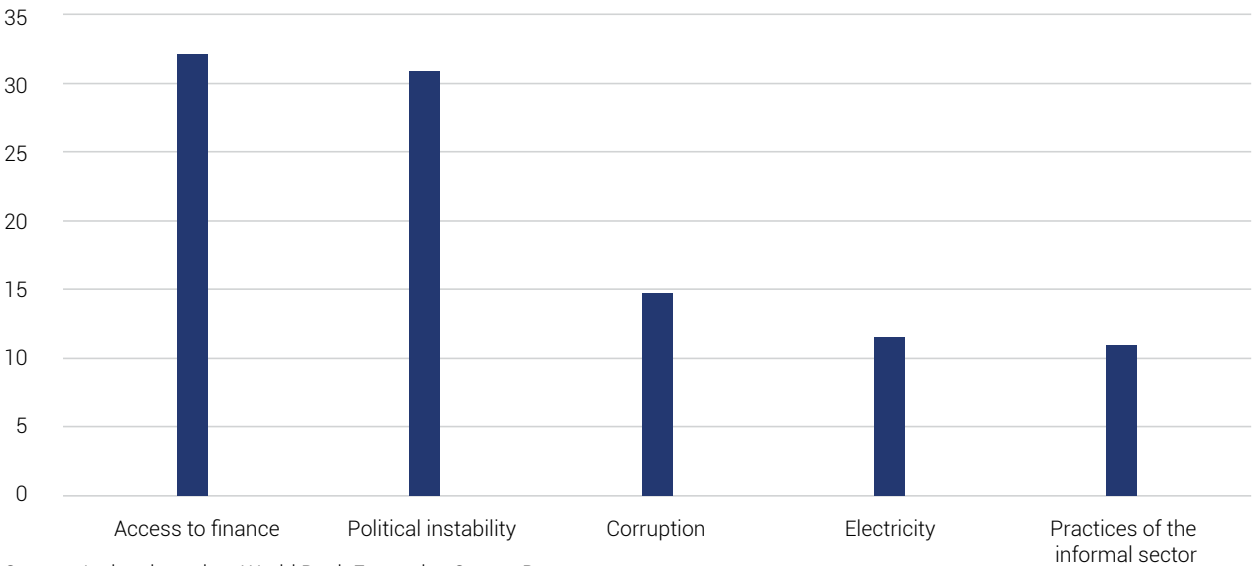
The majority of enterprises in West Africa are in the very-small, small, and medium size categories and constitute an important component of the private sector. It is estimated that about 95% of total enterprises fall under this category and are a major source of income and employment in the subregion. These enterprises can be found in almost all productive and service sectors. They have flexible structures compared to the large-scale enterprises and firms. They tend to be informal and have little or no barriers to start up. This allows women to combine domestic responsibilities and economically productive activities. In general, very small, small, and medium enterprises face several constraints that have implications for their long-term survival. Thus, they face high rates of failure, especially during the first five years.

African SMEs face various major challenges. Top on the list are: (i) lack of access to finance; (ii) poor access to



electricity supply; (iii) poor management; (iv) lack of competency and capability; (v) negative perceptions; (vi) lack of access to reliable information; (vii) limited government support; and (viii) corruption. Other challenges include political instability, labor issues due to the lack of established legal frameworks, lack of coordination, ethnic violence, destruction, and lack of qualified personnel (Bowen et al., 2009; Katua, 2014). A further review of these constraints specific to West Africa, using the World Bank Enterprise Survey data, identified the five largest obstacles to businesses in West Africa in order of significance as follows: (i) lack of access to finance; (ii) political instability; (iii) corruption; (iv) poor access to electricity; and (v) practices of the informal sector (Figure 6).

Figure 6: Largest Obstacles to Businesses in West Africa



Source: Author, based on World Bank Enterprise Survey Data

2.2 Importance of Technology to WSMEs and WVSEs

Rice postproduction technologies are important to WVSEs and WSMEs in West Africa, thus there are compelling reasons to promote women's access to such technologies. Improving access of WVSEs and WSMEs to rice postproduction technologies provides an opportunity to enhance their competitiveness and growth, advance women empowerment, reduce drudgery, and close the gender gap in rice postproduction. Improving access to postproduction technologies also improves rice quality. Furthermore, it increases profitability and contributes to the expansion of the market for rice produced and processed locally. Therefore, improving access of WVSEs and WSMEs to postproduction technologies will lead to the following advantages:

Reduce drudgery on Women. Women in West Africa have low access to rice postproduction technologies, particularly mechanized technologies. Therefore, improving WVSE and WSME access to mechanized technologies will not only reduce drudgery on women (women perform most postharvest activities), and improve their health, but it can also catalyze changes and improvements in rice postproduction activities, including the type and quality of work performed by women.



> **The lower access and the wide gap in the use of mechanized and digital technologies by women can be attributed to impeded access to capital and complementary inputs and services, lower literacy levels, physical accessibility and, in some contexts, gender-biased sociocultural norms.**

Advance women empowerment. In the absence of mechanized technologies, women perform most postproduction activities. However, the role of men becomes more prominent the more the operations become mechanized, thus significantly increasing the reliance of WVSEs and WSMEs on men for operation, management, and repair, with no women involved in most cases. Therefore, the more WVSEs and WSMEs are capacitated to access and take on those roles, the more empowered will women be, as that reduces their dependence on men for labor and improves the jobs created for women and their incomes.

Close the gender gap. Although there are benefits associated with access to mechanized and digital technologies for WVSEs and WSMEs in the rice postproduction, men-owned enterprises have more access. The lower access and the wide gap in the use of mechanized and digital technologies by women can be attributed to impeded access to capital and complementary inputs and services, lower literacy levels, physical accessibility and, in some contexts, gender-biased sociocultural norms. Thus, improvements in the conditions that would lead to increased access by WVSEs and WSMEs will potentially contribute to closing the gender gap observed in rice postproduction in West Africa.

2.3 Taxonomy of Women and Rice Postproduction Enterprises

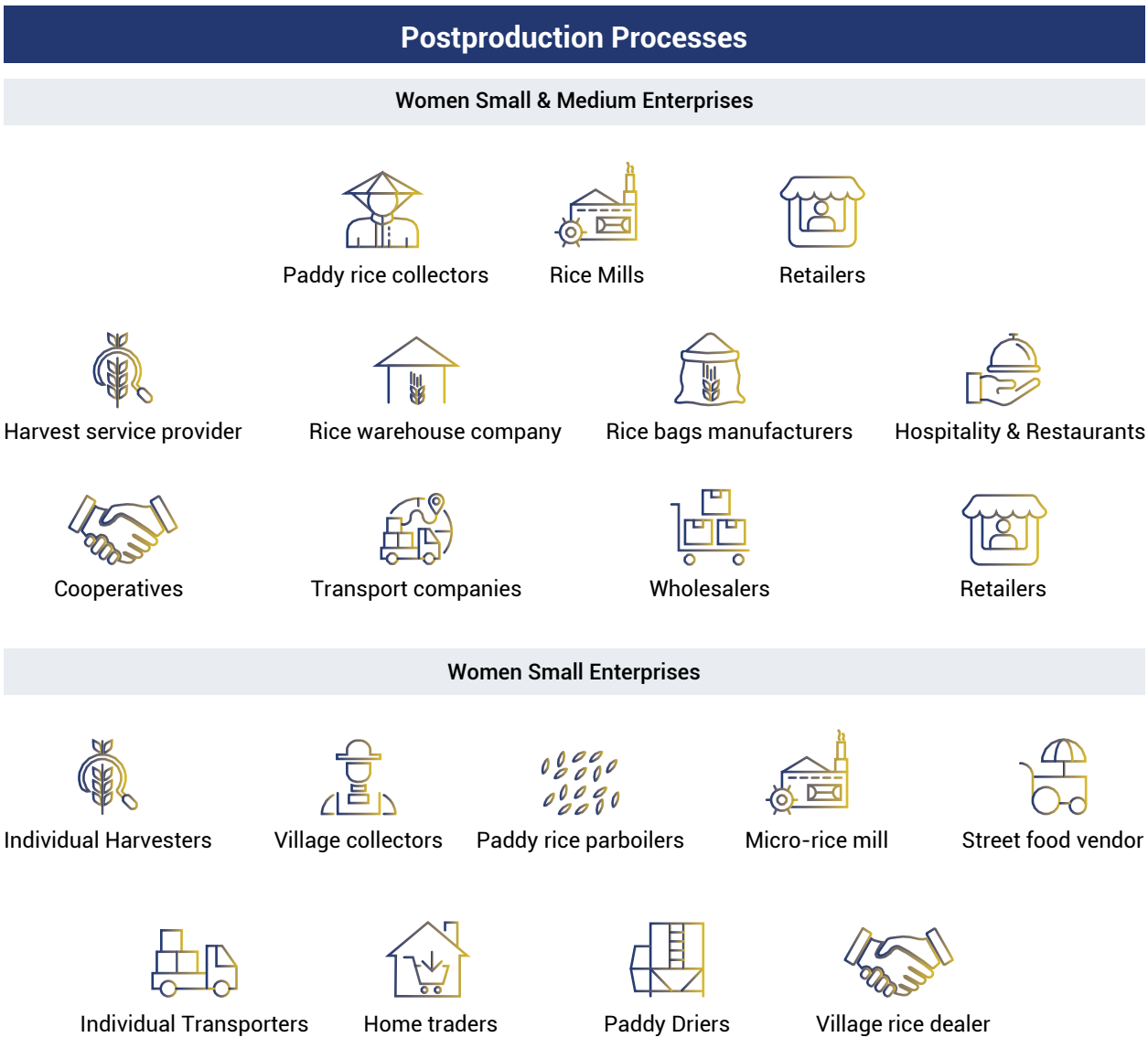
2.3.1 Taxonomy of WVSE and WSMEs

Figure 7 provides a taxonomy of WVSEs and WMSEs in rice postproduction. WVSEs are dominant in the lowest stages of the postproduction processes, with individual business ownership structures.

Though formal, decisions are largely made by the proprietors, and such businesses are much riskier than WSMEs, which tend to be more formalized with a board that meets regularly to review and approve investment plans. WSMEs are more prominent in the postproduction process transition towards higher and modern technologies and markets. It is at this stage that WSMEs could play a bigger role in connecting different components along the rice value chain.

In addition, as rice postproduction processes, especially milling, become more commercialized and industrialized, SMEs play greater roles in linking other segments of the value chain from production and processing to market. However, the prominence of women ownership declines as businesses become larger, more formalized, and use more sophisticated technologies.

Figure 7: Taxonomy of WVSEs and WSMEs



Source: Author

2.3.2 Postproduction Processes and Technologies

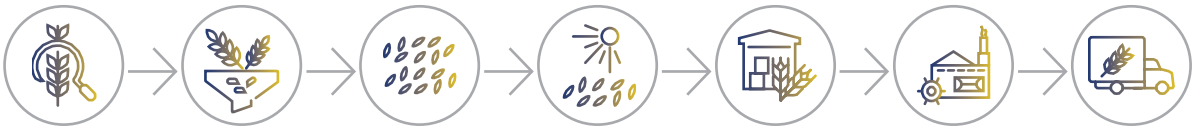
Rice postproduction processes and technologies applied for each process are outlined in Figure 8. In general, there are two categories of rice postharvest technologies: traditional and mechanized. While traditional technologies are labor intensive and do not require much know-how, mechanical technologies require some technical knowledge and skill. In West Africa, most WVSEs and WSMEs are at the bottom of the technology ladder, meaning they mostly use traditional technology, while their male counterparts use mechanized technologies.

Harvesting: At maturity, the mature rice crop is cut and ripped using traditional methods. The use of motorized rice harvesters is not common among WVSEs and WSMEs in West Africa because the rice plots are mostly small and scattered. The harvesting process involves the separation of paddy rice from the panicles. This process is mostly done by women and is labor intensive. However, with the introduction of mechanized land preparation and planting, especially in lowland agro-ecologies, with support from the government and development partners, the need for mechanized harvesters is increasing.

The ideal mechanical rice harvester for small enterprises is the rice ripper. This type of machine can be hand-driven or mounted on the front of a tractor to cut or reap and gather the matured rice. It is optimal in areas with shortages in rice-harvesting labor, though its use in waterlogged conditions is difficult and a major disadvantage.

The use of combine harvesters can be found in the irrigated schemes or lowland agro-ecologies with large scale rice farms spanning over 2,000 hectares of rice field. A combine harvester is a motorized harvesting machine that combines the operations of rice-cutting, with an automatic feeder into a compartment for threshing, cleaning, and discharging of the paddy grain into a bulk wagon at the back or directly into bags, while the thrash is discharged at the other end of the machine. Though the combined harvester is considered the most efficient harvesting (and threshing) machine, its cost is high and it is not suitable for WVSEs and WSMEs. Additionally, it cannot be used in waterlogged and muddy conditions.

Figure 8: Rice postproduction processes and technologies

Process						
						
Harvesting	**Threshing**	**Parboining**	**Drying**	**Storage**	**Milling/ Processing**	**Marketing Wholesale/ Retail**
Process						
• **Traditional** - Sickles - Knives • **Mechanized** - Manual Ripper • **Combined** - Harvester - Wagon/sacs	• **Traditional** - Sticks - Treadle threshers - Trampling • **Mechanized** - Machine threshers - Mobile - Stationery	• **Traditional** - Sun - Mats - Pavement • **Mechanized** - Heat blower . Batch . Flow . Solar	• **Traditional** - Sun - Mats - Pavement • **Mechanized** - Heat blower . Batch . Flow . Solar	• **Traditional** - Storage - Home - Bags - Jute - Woven plastic	• **Traditional** - Mortar & pestle • **Mechanized** - Micro-mill - Two stage mill - Modern rice mill - Miller & Destoner Polisher/ Whitener - Grader, Scale - Bagging machine	• **Traditional** - Cup, Bowls, Tin (various sizes) - Scale - Bags - **Shed** - **Store** - **Truck** - **Supermarket**

Source: Author

Threshing: Traditional threshing, using sticks and sacks, is the most common among WVSEs and WSMEs. Motorized threshing is done using fixed location (stationary), mobile, and combined threshing machines. Over the years, the International Rice Research Institute (IRRI) in the Philippines has developed three main types of stationary threshing machines – one of them being known as the "through-flow" thresher. This type allows stalks and ears to pass through the machine. It consists of a threshing device with pegs, teeth or loops, and (in more complex models) a cleaning-winnowing mechanism based on shakers, sieves, and a centrifugal fan. The second type developed by IRRI in the 70s is the axial flow thresher. This type can be manufactured locally. A more recent addition is a small mobile thresher provided with either one or two threshers. This machine has been widely adopted in many rice-growing areas (Policarpio and Mannamy, 1978) and can also be manufactured locally. The simple design and work rates of these machines seem to meet the requirements of rural communities (FAO 1995).

Drying of Paddy: Drying reduces the humidity level of paddy rice, which must be at least between 12% and 14% for optimal milling and maximum yield, according to FAO. Sun drying is the most common process in West Africa. This involves spreading the paddy on drying surfaces, which could be concrete, tarpaulin, or mats, or on the ground. Impurities like stones, pebbles, seeds are common during this process of drying. The most important financing need of WVSEs and WSMEs at this segment of the value chain is for the purchase of tarpaulins or the construction of concrete drying floors.

Storage: Inadequate access to conducive and milled rice storage is a major constraint to WVSEs and WSMEs in West Africa. Paddy is mostly stored in sacks (jute or plastic) in the open and covered with tarpaulins or in rooms within residential facilities. There are few commercial storage facilities for paddy available to WVSEs and WSMEs in most of the rice-producing areas in West Africa. Those WVSEs and WSMEs with warehouses financed the construction of the facilities from own funds. Poor ventilation and the lack of humidity control, rodents, and pest infestation (especially weevils) degrade the quality of the stored paddy rice, leading to significant losses.

Rice processing: Rice processing plays a very important role in the postproduction segment of the rice value chain. Traditional and village-level micro-mills are most common in West Africa. This is a major constraint contributing to degradation in the quality of domestic rice. In West Africa,



there are three types of rice milling systems: (i) one step – in which the husk and bran are done in a single process to get the milled or white rice; (ii) the removal of husk and bran are done separately, resulting in the production of brown rice as an intermediate product; and (iii) multi-stage milling, which involves different processes. Depending on size and location, two categories of rice mills are available: (i) small-scale or mini rice mills (at the village level); and (ii) commercial rice mills.

While WVSEs rely largely on paying a service provider to mill their paddy, some WSMEs aspire to own small or mini rice mills. A rice mill comprises various components, such as the mill machine, cleaner, de-stoner, etc. Thus, the total cost of setting up a rice processing plant depends on several factors, including: (i) the type of processing technology (hammer mill, complete modern rice mill plant, etc.); (ii) the capacity of the rice milling plant; (iii) the cost of land and construction; (iv) the availability of electricity and cost of energy; and (v) operating costs.

Paddy parboiling: Paddy parboiling is an important process in adding value to milled rice and is exclusively done by women. The process involves soaking and steaming prior to drying. The soaking and steaming process normally takes two days. Apart from being a labor-intensive operation, it uses a lot of firewood and water. Local manufacturers produce simpler, less efficient parboilers that consist of a drum of standard capacity of about 200 liters, fireplace, and water drums/containers for soaking the paddy rice overnight. However, there are improved parboiling designs that can be produced locally. The newest technology is the innovative and environmentally friendly parboiling system called "Grain quality enhancer, Energy-efficient and durable Material (GEM)" designed with support from AfDB and Africa Rice. GEM is designed to meet the needs of WVSEs and WSMEs. However, they require financing to meet the financial outlay for its construction and installation.

Marketing: Traders are the link between paddy processors and consumers. They ensure the provision of rice to the country's main urban and rural markets. The main actors in this segment are the wholesalers and the retailers. There is also a remarkable presence of women traders, especially in the small-scale (mi) processing level.

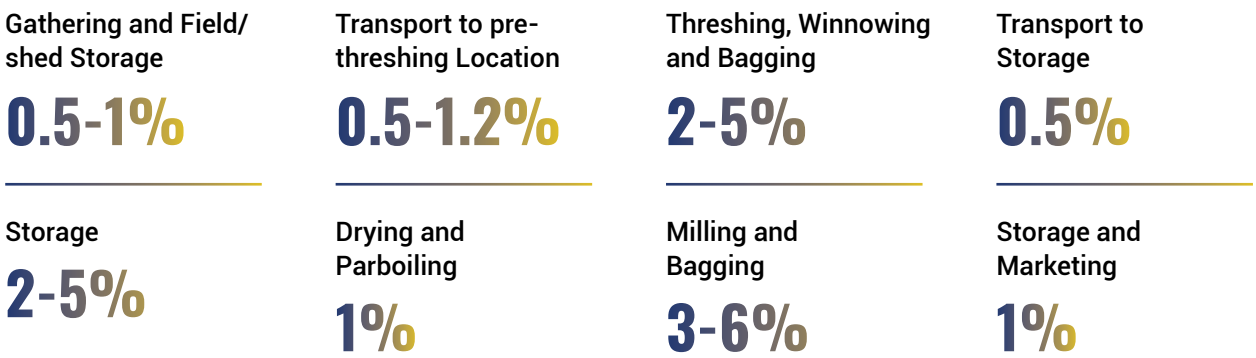
Transport: Transportation is particularly required for aggregators of paddy or milled rice to transport their produce to the market for bulk sales. Among WVSEs, hiring of transport – motorcycles or trucks, or paying for space in commercial transport, are common. Private operators generally provide entrepreneurs with transportation services. In the Senegal River valley, some river transportation is provided by private operators to transport milled rice to market.

Transportation is particularly required for aggregators of paddy or milled rice to transport their produce to the market for bulk sales. Among WVSEs, hiring of transport – motorcycles or trucks, or paying for space in commercial transport, are common.

Postproduction losses: A schematic presentation of rice postproduction activities and associated losses among WVSEs and WSMEs is presented in Figure 9. Cumulatively, WVSEs and WSMEs lose approximately 10.5% - 22% of the rice during the postproduction phase, thus negatively affecting incomes. These losses are associated with the financial and non-financial constraints faced by WVSEs and WSMEs.

As a consequence of limited financial capacity, WVSEs and WSMEs largely use manual labor (mostly women) and rudimentary machines and equipment to perform postproduction activities, thus leading to high postproduction losses. The drudgery of these manual activities affects the health of women, especially in rural areas with limited and poor access to health facilities.

Figure 9: Rice Postproduction Activities and Losses (%)



Source: Author, based on field interviews in Sierra Leone 2021

2.4 WVSE and WSME Constraints to Access Technology

2.4.1 Why access to technologies matters

WVSEs and WSMEs play vital roles in the rice postharvest value chain in West Africa. Improving their access to technologies provides an opportunity to enhance their competitiveness and growth, advance women empowerment, reduce drudgery, and close the gender gap in rice postproduction. Access to rice postproduction technologies is also critical for improving rice quality and profitability and expanding the market for rice produced and processed locally.

WVSEs and WSMEs active in rice postproduction in West Africa have limited access to technologies, especially machines and equipment, due to financial and non-financial barriers, including: (i) limited access to finance for mechanization; (ii) inadequate awareness of the operation and maintenance of postproduction mechanization facilities; (iii) low level of technical knowledge and skills; (iv) lack of women-friendly technologies; mechanized technologies and equipment on the market are not attractive for adoption by women entrepreneurs.

Among these barriers, the most compelling is access to finance. Accessing technology requires financial outlays that are not readily available to WVSEs and WSMES. Therefore, difficulty to access finance directly limits the ability of WVSEs and WSMEs to leverage postproduction rice technologies to increase profits and grow their businesses. It has been established that WVSEs and WSMEs in rice postproduction often encounter difficulties to acquire bank loans, venture capital, investment, and other business opportunities. They are also adversely affected by discriminatory lending practices, lack of collateral, and gender bias in financial institutions. Specifically, improving access for WVSEs and WSMEs to postproduction technologies will lead to the following advantages:

Reduce drudgery. Women in West Africa have low access to rice postproduction technologies, particularly mechanized technologies. Therefore, improving WVSE and WSME access to mechanized technologies will not only reduce drudgery on women (women perform most postharvest activities) and improve their health, but can also catalyze changes and improvements in rice postproduction activities, including the type and quality of work performed by women.



Advance women empowerment: In the absence of mechanized technologies, women perform most postproduction activities. However, the role of men becomes more prominent as the operations become more mechanized. Thus, WVSE and WSME reliance on men for operation, management, and repair increases significantly, with no woman involved in most cases. Therefore, the more WVSEs and WSMEs are capacitated to access and take on those roles, the more women will be empowered, because this reduces their dependence on men for labor and improves jobs created for women and their incomes.

Close the gender gap: Although there are benefits associated with access to mechanized and digital technologies for WVSEs and WSMEs in the rice postproduction, men-owned enterprises enjoy more access. The lower access to and wide gap in the use of mechanized and digital technologies by women can be attributed to barriers to access capital and complementary inputs and services, lower literacy levels, physical accessibility, and, in some contexts, gender-biased sociocultural norms. Thus, improving on such conditions will potentially contribute to closing the gender gap observed in rice postproduction in West Africa.

Address gender information asymmetries: WVSEs and WSMEs active in rice postproduction in the four study countries indicated that women had fewer opportunities, compared to their male counterparts, to receive adequate information and knowledge on improved postproduction opportunities. There is insufficient information about types of available female-adapted technologies. Therefore, in addition to financial and non-financial inhibitors, information asymmetry affects the ability to adopt and use such technologies.

Address design problems: Despite the dominance of women in postproduction activities and the associated drudgery, the design of rice postproduction machinery and tools – for example the rice ripper used for rice harvest - continue to be designed primarily to suit male workers. The complexity and the physical strength needed to operate such machines put women at a disadvantage, as they have to contend with additional challenges in hiring and supervising male workers, even where they have access to the machines and equipment.

Address the gender digital literacy gap: The gender digital gap refers to disparities between men and women in internet access, digital skills, and usage of technologies. Studies on this topic have shown that women are less likely to have access to smartphones, computers, and the internet, which hinders their participation in online business activities and e-commerce. Key reasons for this include deep-rooted gender stereotypes and cultural norms that often dictate traditional gender roles, relegating women to domestic responsibilities rather than entrepreneurial pursuits. Also, societal expectations may discourage women from pursuing business ventures or accessing formal education and training opportunities. In general, cultural norms that prioritize male-dominated industries or discourage women from taking on leadership roles can hinder women's participation in SMEs.

Access to smartphones by WVSEs and WSMEs is becoming significantly high in West Africa. While access to mobile phones is increasing because of individual local mobile phone service providers in rural areas, ownership of smartphones by women VSE and SMEs is low. Relatedly, the use of digital financial services by WVSEs and WSMEs, for example electronic banking services and platforms, is still minimal. Low financial literacy, poor knowledge, poor infrastructure, and unaffordability of internet services are key reasons for the limited use of digital financial services.

2.4.2 Obstacles to access technologies

WVSEs and WSMEs face barriers that limit their ability to leverage technology for business growth and innovation. In this study we focused on: (i) mechanical



WVSEs and WSMEs active in rice postproduction in the four study countries indicated that women had fewer opportunities, compared to their male counterparts, to receive adequate information and knowledge on improved postproduction opportunities.





technologies (access to machines and equipment); and (ii) digital technologies (ICT), especially the disparities in internet access, digital skills, and technology usage by women entrepreneurs. The barriers to access of women entrepreneurs to these technologies are discussed in following paragraphs.

Difficulty of access to finance for rice postproduction technologies: Accessing technology requires financial outlays that are not readily available to WVSEs and WSMEs. Therefore, the difficulty of access to finance directly limits the ability of WVSEs and WSMEs to leverage postproduction rice technologies to increase profits and grow their businesses. It has been established that WVSEs and WSMEs in rice postproduction often encounter difficulties to acquire bank loans, venture capital, investments, and other business opportunities. They are also adversely affected by discriminatory lending practices, lack of collateral, and gender bias in financial institutions.

Technical and knowledge and skills: WVSEs and WSMEs suffer technical constraints and knowledge gaps, which are attributable to: (i) inadequate awareness of available rice postproduction technologies, (especially mechanization); (ii) inadequate knowledge of improved rice postproduction machines and equipment; (iii) poor capacity of national research and extension services to develop and disseminate relevant technology needs of women; and (iv) Insufficient information about types of available women-friendly technologies.

Inadequate women-friendly technologies: Available postproduction technologies are not only inadequate, but they are also not women-friendly. This is attributable to the limited attention given to the needs of women, and the physical disparities between men and women.

Insufficient skill: In general, women in rice postproduction have limited skills to operate, maintain, and manage machines and equipment for performing various tasks to realize efficiency and grow their businesses. Also, the low level of literacy of women entrepreneurs in the rice postproduction value chain compromises their ability and capacity to understand, appreciate, and use technologies in their businesses in general. Thus, women entrepreneurs rely on male technicians to operate and manage such machines and equipment.

Market barriers. The availability and prices of rice postproduction machines and equipment pose barriers to WVSEs and WSMEs. Rice postproduction technologies are either imported or have to be manufactured locally on demand, thus the costs associated with acquiring machines and equipment are mostly unaffordable, leaving WVSEs and WSMEs reliant on government and donor-financed projects and NGOs to provide access, through grants and/or loans.

Limited infrastructure: WVSEs and WSMEs, like their male counterparts, are constrained by poor infrastructure (especially internet and electricity) in rural areas. Sustainable and affordable access to electricity and the internet encourages the use of digital technologies. However, electricity is not always available in rural areas. Also, the cost of internet access directly hinders access and use of digital technologies in rice postproduction businesses.

Sociocultural norms: There are deep-rooted gender stereotypes, often imposing traditional gender roles in West Africa. Some of these cultural norms relegate women to domestic responsibilities, which in turn affects the availability of time to invest in acquiring knowledge and skills, as well as accessing formal education and training opportunities to adequately prepare them to operate, manage, and maintain technology assets.

Training gaps to access technologies: WVSEs and WSMEs in rice postproduction rely on family, relatives, and friends to provide labor. These enterprises have difficulty to attract and retain skilled technicians and other vital employees, as they tend to offer less appealing remuneration and working conditions. While training and upskilling women in the rice postproduction value chain is important for empowering women entrepreneurs, due to limited number of employees, they often lack the internal capacity to organize training related to technologies for business improvement. Table 3 ranks the five most important training needs of WVSEs and WSMEs based on interviews conducted in Sierra Leone.



WVSEs and WSMEs in rice postproduction rely on family, relatives, and friends to provide labor. These enterprises have difficulty to attract and retain skilled technicians and other vital employees, as they tend to offer less appealing remuneration and working conditions.

Table 3: Ranking of training needs of WVSE and WSMEs

Training Needs of WVSEs and WSMEs	Rank
Operation of Postproduction machines a) Threshers and winnowers) b) Small-scale rice mills)	1
Accessing loans and loan management	2
Postproduction/postharvest management	3
Group management and leadership	4
Business skills	5

Source: Sierra Leone Country Report

The most crucial training needs for the operation of postproduction machines are: (i) use and operation of threshers and winnowing machines; and (ii) use and operation of small-scale rice milling machines. Because machines are traditionally operated by men, women are usually not included in the training organized by suppliers when machines are installed. As postproduction activities in the rice value chain are dominated by women, it is only logical that women should be trained on the use, operation, maintenance, and management of postproduction machines.

Most WVSE and WSME managers/owners are not well informed about the processes and procedures for accessing loans, particularly from commercial banks. They are also not trained on book management, simple accounting, savings, and loan management. Governments in the West African countries provide support to WVSEs and WSMEs through relevant ministries, departments, agencies, or financial institutions. However, the low level of financial literacy among women entrepreneurs in general poses additional constraints to their access to loans. Those who do access such loans tend to default due to several other reasons. In this respect, WVSEs and WSMEs require specific training on the documentation required by commercial banks/microfinance institutions, and the procedures for access to loans, bookkeeping, savings, loan use and repayment planning, etc.

Postproduction management is rated as the third most important area for training. This is the period when most activities of the WVSEs and WSMEs take place. Women entrepreneurs in the postproduction rice value chain need to be trained on all operations and their interconnections for competitiveness. In addition to training on key postproduction activities, training is also required on logistics, costing, and rice quality management.



Group management and leadership training is the fourth priority area listed by WVSEs and WSMEs. Training on identifying perceived needs, group formation, group dynamics, group management (including business registration), and leadership skills are some areas identified for training.

Training on business skills for women is the fifth priority area identified. Training in this area could include the definition of a business and how to succeed as a woman entrepreneur in starting a new business, setting up business goals and success factors, negotiations skills, growing existing business, and opportunities for women entrepreneurs.

2.4.3 Implications of poor access to technologies for WVSEs and WSMEs

Although there are more WVSEs and WSMEs in rice postproduction than their male counterparts, the number diminishes as the size of enterprise increases and with the required ownership of postproduction technologies. A review of data from West Africa showed that the larger the business and sophistication of the owned rice postproduction technology owned, the lower the share of WVSEs and WSMEs gets. It is known that more value is accrued along the rice postproduction value chain from harvest to milling and sale of milled rice. Also, it is observed from the four country studies in West Africa that the more value is added along the rice value chain, the fewer involved WVSEs and WSMEs are. Lack of access by WVSEs and WSMEs to technology has negative impacts on the profits, competitiveness, and growth of the enterprise.



Effect on improvements in efficiency and business processes: Mechanization is known to improve efficiency and reduce drudgery in rice postproduction activities. In the case of digital technology, key information surveys on WVSEs and WSMEs revealed that women entrepreneurs accessing digital technologies, especially mobile phone, had better information about their businesses and the wider market network. Thus, without access to technology, WVSEs and WSMEs will not be able to improve efficiency, reduce drudgery, and access business information to improve market network.

Financial instability: Rice postproduction activities, like threshing, milling, and parboiling, are labor-intensive and require hard work. In the case of parboiling, lack of access to water- and fuel-efficient technologies leads to the use of more water and fuel, mostly fuel wood, thus increasing cost. Therefore, without access to labor-saving water- and energy-efficient technologies, WVSEs and WSMEs will need more financial resources to pay for labor, water, and fuel, thus reducing profitability and contributing to financial instability, as well as deforestation. The lack of access to digital business technologies leads to financial dependence and increased travel expenses, which could have otherwise been saved through digital money transfers.

Limited market and business network: Access to digital business technologies, especially social media, is known to transform and expand marketing opportunities, as WVSEs and WSMEs that use social media have a better chance to access business information and a wider network to market opportunities, and to maintain quality relationships with customers.

Quality and price of rice: Quality of rice is a major factor used by consumers when procuring and consuming rice. Lack of access to postproduction technologies, especially harvesting and threshing mills, leads to poor quality of rice from WVSEs and WSMEs. This translates to lower prices compared to imported higher quality rice, thus affecting profitability and growth of businesses.





03

Recommendation and **Conclusions**

3.1 Recommendations

There is a range of barriers and factors that constrain WVSE and WSME access to rice postproduction technologies. These include lack of finance or financing; lack of information, knowledge, and skills; limited women-friendly technologies; market barriers; limited or poor infrastructure; and sociocultural norms. In that context, following are recommendations to enhance WVSE and WSME access and use of rice postproduction technologies.

A holistic approach to supporting WVSEs and WSMEs in rice postproduction. WVSEs and WSMEs in rice postproduction face endemic gender inequality and are constrained by multiple and interacting barriers that hinder business growth. In this respect, a unidimensional approach, such as improving access to finance through facilitating affordable loans, grants, cash transfers, or training has not proven to be effective and sustainable. As access to technology, finance, and knowledge are closely interrelated, addressing any of them alone will not solve the constraints binding WVSEs and WSMEs. Therefore, it is recommended that holistic rice postproduction value chain development emphasizing an ecosystems approach be adopted to address the constraints associated with harvesting and threshing, parboiling and drying, and milling and marketing.

Gender-responsive policies for WVSEs and WSMEs. Issues relating to WVSEs and WSMEs in rice postproduction should not be decoupled from the overall policies and strategies by governments to address gender barriers and women economic empowerment. In this regard, governments and development partners supporting WVSEs and





WSMEs should strengthen policies and initiatives to promote women's access to finance, technology, and business support services. This requires a multistakeholder approach through which policymakers in West Africa would strengthen the implementation of laws and regulations that promote active participation of women in the economy, especially as agro-entrepreneurs, by ensuring equal rights to property ownership, inheritance, and access to justice for women entrepreneurs. To improve financial access, governments and financial institutions should implement inclusive financing policies that facilitate women's access to credit, loans, and financial services. This may involve establishing dedicated loan programs, providing subsidies or guarantees, and promoting financial literacy among women entrepreneurs.

Strengthen WVSEs and WSMEs. There are multilateral development agencies, organizations, and platforms with support from governments and development partners that address gender disparities and women economic empowerment in West Africa. The capacity of such agencies, organizations, and platforms to effectively address gender disparities in general, and economic empowerment of WVSEs and WSMEs in rice postproduction in particular, should be integral to the holistic approach. Therefore, supporting WVSEs and WSMEs in rice postproduction should include strengthening women's business associations and other similar civil society organizations. Such capacity strengthening should include technical and business support and advice, policy and advocacy to public sector development partners, and approaches and methods to foster competitiveness and growth of women's entrepreneurship.

Improve milled rice quality. There is growing demand for good quality parboiled rice in West Africa. However, the traditional and common parboiling process requires lots of firewood or charcoal and water, which is not only inefficient, but also laborious, time-consuming, and unsafe. The process also produces low quality rice with impurities, broken and burnt grains, and undesirable

5. https://www.africarice.org/gem-rice-parboiler

odors. As part of the effort to improve rice quality and energy efficiency in rice parboiling, Africa Rice and its partners have developed an improved rice parboiling technology called 'Grain quality enhancer, Energy-efficient and durable Material,' (GEM). This improved parboiling technology produces rice of high physical and edible quality compared with the traditional technology[5]. It is more efficient, economic and timesaving than the traditional parboiling process. The GEM technology should be promoted as part of the holistic approach to addressing rice postproduction constraints.

Bridge the information gap. WVSEs and WSMEs in rice postproduction in West Africa often lack adequate information on labor-saving and efficiency technologies, including digital business technologies. They are also constrained by limited data and knowledge to estimate the benefits of the technological upgrading of their businesses. Governments and development partners, including NGOs, in the major rice producing countries in West Africa are active in providing targeted support to WVSEs and WSMEs. However, while the barriers are often known, government and development partner-funded rice value chain development projects often focus on donating machines and equipment to individuals or rice farming communities on a no-cost basis, without putting in place mechanisms to assure sustainability. While this approach considers the limited financial capability of WVSEs and WSMEs in rice postproduction, it has not proved to be sustainable and beneficial to a large number of women. However, new approaches that require enterprises to co-finance the purchase of rice postproduction technologies at lower-than-market prices for commercial banks have proven to be more sustainable.

Support training for SMEs to access and use Digital Business Technologies. WVSEs and WSMEs in rice postproduction value chain in West Africa are constrained by infrastructure limitation and limited resources and expertise to effectively access digital business technologies, especially the internet, internet-based software for financial management, mobile banking, etc. Although governments are increasingly focusing on promoting access to the internet, its use among small enterprises remains low. Thus, there is a need for governments to partner with NGOs to provide digital-skills training programs tailored to the needs of women entrepreneurs.

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3.2 Conclusion

Recommendations provided by this study should not be implemented in isolation. Also, noting the pervasive gender gaps in West Africa, especially in rural areas, special initiatives for women entrepreneurs are required to facilitate accessibility, affordability, and utilization of rice postproduction technologies. However, while a holistic approach is required to address barriers to WVSE and WSME access to technology and use, it must be done within the national gender and women economic empowerment policies.

In general, acquiring rice postproduction technologies requires high investments, which poses a major barrier to WVSEs and WSMEs, as they are usually constrained by limited budget resources. In this respect, governments and development partners, including financial institutions, should combine financial incentives with technical assistance, subsidies, grants, or low-interest loans to WVSEs and WSMEs for purchasing or renting rice postproduction machinery and equipment or for paying providers for such services.

Access by WVSEs and WSMEs to rice postproduction technologies can reduce drudgery on women, improve the physical quality of rice, and save time that can be used for household and/or other productive activities. However, the lack of technical skills and knowledge and the complexity of new rice postproduction machines and equipment and inability to operate and maintain them effectively may be additional factors that inhibit access and utilization. Therefore, governments and development partners, in collaboration with rice postproduction machine and equipment suppliers, should organize promotional campaigns and training for WVSEs and WSMEs in rice postproduction.

Improving access for WVSEs and WSMEs in rice postproduction to digital technologies should also be done within the framework of improving the national digital infrastructure. At the enterprise level, there are basic conditions that need to be met. These include availability and affordability of the technology, internet connectivity and stability, and availability of trained and skilled ICT staff. Equally important is the availability, stability, and affordability of electricity to power digital equipment and businesses in particular.

Limitation of the study. While this study has largely focused on access by WVSEs and WSMEs to rice postproduction technologies, it has not assessed the profitability of rice postproduction technologies or the conditions under which such technologies could be profitable or not profitable. However, it is a known fact that profitability of technology investment, especially by VSEs and SMEs, is dependent on several factors, including appropriateness of the technology, location, market demand, capacity, utilization per annum, incentives, training and skills, and product market.



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